



Cesar M. Manna, Ph.D.

R&D Director | Engineering & Technology Leader



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About me

Cross-industry R&D and engineering leader with **15+ years** driving complex, multidisciplinary technical programs across materials, devices, and system integration. Proven track record of **engineering execution**, root cause analysis, and translating deep technical insight into commercial products. Led cross-functional teams and multidisciplinary organizations, owning a **\$1.2M annual budget**. Named inventor on **6 granted patents**. Experienced in **executive-level communication**, technology roadmap definition, and operating across the full product lifecycle, from R&D through commercial introduction. Direct domain experience in **digital PCB and advanced electronics manufacturing** (Nano Dimension).

Patents

Named inventor on 6 granted patents across additive manufacturing and materials engineering (US10589459B2, US10994475B2, US10328635B1, US11390016B2, US12070908B2, WO2020144678A3) and 3 pending applications (WO2025116947A1, US20260008230A1, GB202414384D0) spanning Stratasys, Massivit, Nano Dimension, and PV Nano Cell.

Experience

2023–Today **PV Nano Cell, Renewable Energy**

VP R&D

Early-stage solar materials company; cross-functional R&D and production leadership, \$1.2M annual budget, team grown from 6 to 10. Defined 2-year technology roadmap.

- Lead complex, multidisciplinary technical investigations spanning **material, mechanical, functional, and system-level performance**; drive engineering improvements based on data-driven root cause analysis.
- Define and enforce R&D standards, testing methodologies, and KPI-driven cadence with strategic industry partners.
- Serve as primary technical authority across the full product lifecycle, from early-stage development through commercial introduction; present technology strategy to board and executive stakeholders.
- Conduct direct customer discovery with solar market leaders; translate findings into quarterly R&D priorities aligned with CEO, ensuring development is **market-pulled**, not technology-pushed.
- Manage and mentor R&D and production teams through weekly cross-functional meetings, eliminating bottlenecks between discovery and scale-up.
- Oversee IP strategy; filed patents protecting core material innovations.

2020–2023 **Nano Dimension, Digital PCB** *Director, Materials Discipline Manager*

*Publicly traded deep-tech company (NNDM) in **digital PCB additive manufacturing**; grew R&D team from 4 to 10, \$1.2M annual budget.*

- Led multidisciplinary analytical and engineering teams across **10 researchers**; provided mentoring, prioritization, and performance oversight across exploratory and applied workstreams in **PCB materials and process engineering**.
- Partnered with Product Management and Systems Engineering on technical roadmap direction, feasibility assessment, and milestone execution; drove cross-functional alignment between R&D, production, and commercial teams.
- Introduced bi-weekly R&D and product alignment meetings, coupling development cycles directly to commercial milestones and cross-functional initiatives.
- Launched weekly literature review program that generated a new materials patent, the **first filed in 5+ years**, demonstrating data-driven innovation leadership.
- Implemented monthly Agile sprints, improving milestone predictability and cross-team visibility.
- Designed company-wide innovation competitions, accelerating product pipeline ideation.

2021–2022

R&D Materials Team Leader

- Led development and scale-up of **dielectric ink technology** for PCB commercial production; conducted root cause analysis to improve product quality and manufacturing efficiency.
- Facilitated close collaboration between R&D and production teams, enhancing material performance characterization and reducing scale-up risk.



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Languages

English – C1 Proficient

Hebrew – C1 Proficient

Spanish – Native

2020–2021

R&D Senior Chemist

- Developed and commercialized a high-performance **dielectric ink** for PCB inkjet printing, improving product performance by **30%** and increasing market competitiveness.
- Created ceramic dielectric formulations for inkjet applications via organic synthesis and sol-gel processing, expanding the company's PCB product portfolio.

2018–2020 **Eltra Helion Ltd., Agency & Distribution**

Specialist Sales Manager

- Drove **250%** increase in annual turnover through technical consultancy; developed direct executive-level client relationships and commercial positioning strategy.
- Gained hands-on commercial experience managing pricing strategy and market positioning, complementing deep technical expertise with customer-facing delivery.

2017–2018 **Massivit, 3D Printing Large Scale**

R&D Senior Chemist

- Led development of novel water-soluble support materials for large-scale 3D printing via organic synthesis and photopolymer formulation, creating a new product line.
- Applied UV photopolymerization and **rheology optimization** to tune material properties; conducted mechanical and functional performance characterization for large-format deposition systems.

2015–2017 **Stratasys, 3D Printing**

R&D Senior Chemist

- Developed material formulations that increased product durability by **40%**, reducing field failure rates through systematic root cause analysis and data-driven design decisions.
- Pioneered thermally conductive material systems using inorganic filler dispersion and composite formulation, opening new market segments in advanced manufacturing.
- Characterized materials using **DSC, TGA, and mechanical testing** to guide formulation decisions and meet performance specifications across the full development lifecycle.

2013–2014 **Boston College**

Postdoctoral Associate, Prof. J. A. Byers

- Iron-based catalysts for sustainable poly(lactic acid) synthesis.

2011–2012

Massachusetts Institute of Technology *Postdoctoral Associate, Prof. C. C. Cummins*

- Inorganic and organometallic synthesis; cyclo-phosphate complexes with transition metals.

Education

2006–2011 **Ph.D., Bio-inorganic Chemistry** The Hebrew University of Jerusalem